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1. INTRODUCTION

This section provides an overview of the Software Requirements Specification (SRS) document for the Property Management System (PMS). It outlines the purpose, scope, definitions, and conventions used throughout this document.

1.1 Purpose

This document defines the functional and non-functional requirements for the Property Management System (PMS), developed to manage, monitor, and control government-owned assets across all administrative and organizational units within the Oromia Regional Government.

This document serves as a formal reference for system development, testing, validation, and approval by all stakeholders including developers, project managers, quality assurance teams, and government officials.

1.1.1 Vision Statement

The PMS enables quick and accurate inventory management of government assets. The system facilitates tracking of asset status and current assignments, replacing traditional PIN codes with modern barcode and QR code technology. It enables scanning and printing of barcodes while maintaining comprehensive supplier information, asset disposal records, employee data, and maintenance histories.

The ultimate purpose of PMS is to reduce clerical labor, accelerate workflow processes, and provide greater accuracy in asset management. The system aims to enable public bodies to achieve their objectives through effective and efficient management of their assets, playing a critical role in managerial decision-making. As a public body tool, it provides a comprehensive understanding of asset scope, work unit needs, and departmental expectations.

PMS shall maintain equivalent electronic records of BIN CARD, User Card (UC), Stock Record Card, and other paper-based documentation (Model 19 and Model 22 forms remain in use). This minimizes the effort required to search for information from paper records. An exhaustive history of all transactions shall persist, categorized, and presented through an intuitive navigational interface.

In addition to providing a stable and professional working environment, PMS is expected to deliver:

- Enhanced visibility and increased efficiencies in asset management transactions
- Reduction in spending through asset reuse
- Automated warnings based on asset-related events
- Easier risk prediction and management reporting

- Informed asset investment decisions
- Demonstrated regulatory compliance
- Comprehensive reporting capabilities by various classifications
- Visibility and reuse of assets in storage
- Visibility of asset reliability by manufacturer, model, and user
- Reduction in asset purchase costs through reuse
- Automatically updated asset registers
- Bar-coding and QR-coding for asset tagging with periodic scanning to ensure correct inventory

1.2 Scope

The Property Management System (PMS) shall provide the following capabilities:

- Centrally manage all government-owned properties (assets) across the Oromia Regional Government
- Support hierarchical administrative structures from regional to woreda levels
- Manage stock transactions including Receive, Issue, Transfer, and Dispose operations
- Register and track fixed assets throughout their lifecycle
- Support depreciation calculations for fixed assets
- Enforce administrative-boundary-based data visibility and access control
- Provide hierarchical reporting capabilities from Woreda to Regional levels
- Support Barcode and QR-based asset identification and tracking

Out of Scope:

The system shall not include:

- Full procurement execution workflows
- Financial accounting modules
- Budget management functionality
- Vendor payment processing

1.3 Definitions and Acronyms

The following table defines key terms and acronyms used throughout this document:

Term	Definition
PMS	Property Management System
Address	Administrative location including Region, Zone, City, Sub-city, or Woreda
Organization	Sector office such as Education, Health, or Agriculture
Store	Physical storage unit within an organization

Term	Definition
Fixed Asset	Long-term individually tracked asset with extended useful life
Utility	Consumable or short-term asset
FAR	Fixed Asset Registration
Barcode	Machine-readable code for asset tagging
QR Code	Quick Response Code for asset tagging
MoFED	Ministry of Finance & Economic Development
PIN	Property Identification Number
UC	User Card - Fixed Assets with User Control Card
Issue Request	List of items/properties and quantities requested from departments and/or individuals
Receive (Model 19)	Official form for recording assets received
Issue (Model 22)	Official form for recording assets requested and issued
EFG	Ethiopian Federal Government
PAO	Property Administration Officer
PPO	Procurement Directorate Officer

1.4 Document Conventions

This Software Requirements Specification adheres to the following conventions:

- **Requirement Identifiers:** Functional requirements are prefixed with "FR-" followed by a sequential number (e.g., FR-1, FR-2)
- **Use Case Identifiers:** Use cases are prefixed with "UC-" followed by a sequential number (e.g., UC-01, UC-02)
- **Typography:**
  - **Bold text** indicates emphasis on key terms or important concepts
  - *Italic text* indicates system states, status values, or field names
  - Monospace text indicates system codes or technical identifiers
- **Priority Levels:** Requirements may be classified as:
  - **Critical:** Must be implemented for system operation
  - **High:** Important for system functionality
  - **Medium:** Desirable features
  - **Low:** Optional enhancements
- **Tables:** Used to present structured information such as definitions, roles, and requirements
- **Lists:** Bullet points indicate non-sequential items; numbered lists indicate sequential steps or prioritized items

2. OVERALL DESCRIPTION

This section provides a high-level overview of the Procurement and Property Management System, including its context, major functions, user characteristics, and operational environment.

2.1 System Overview

The Property Management System (PMS) is a centralized web-based application deployed at the regional level utilizing a single PostgreSQL database instance. All administrative addresses access the same system instance, with data isolation enforced logically through administrative hierarchy structures and role-based access control (RBAC) mechanisms.

The system provides comprehensive asset management capabilities including stock management, fixed asset registration, request workflows, and hierarchical reporting. All transactions are audited and maintained in a centralized database to ensure data integrity and traceability.

2.2 Administrative Hierarchy Model

The system shall support the following administrative hierarchy structures:

Hierarchy Pattern 1:

- Country → Region → Zone → Woreda

Hierarchy Pattern 2:

- Country → Region → City → Sub-city → Woreda

Hierarchy Pattern 3:

- Country → City → Sub-city → Woreda

Hierarchy Requirements:

Each administrative address shall:

- Maintain a parent-child relationship with adjacent levels
- Support upward reporting aggregation to higher administrative levels
- Be uniquely identifiable within the system
- Enforce data visibility boundaries based on hierarchical position
- Allow users to access only data within their administrative scope and subordinate units

The system shall prevent circular hierarchy relationships and enforce referential integrity across all administrative levels.

2.3 User Roles and Actors

The following table summarizes the primary user roles and actors within the system:

Actor	Description
Regional Administrator	Full access to all regional data and operations
Zone/City Administrator	Access to zone/city and subordinate unit data
Woreda Administrator	Access to woreda-level data and operations
Director / Organization Head	Approves asset requests from subordinate staff
Storekeeper	Manages stock transactions and inventory operations
Store Clerk	Assists storekeeper with inventory operations
Property Officer (PPO/PAO)	Registers and manages fixed assets
Employee	Submits asset requests and views personal request status
Auditor	Reviews transactions, reports, and compliance

Detailed role definitions and responsibilities are provided in Section 4.

3. SYSTEM FEATURES AND FUNCTIONAL REQUIREMENTS

This section describes the functional requirements of the Procurement and Property Management System, organized by feature area. Each requirement is uniquely identified and describes a specific system capability.

3.1 Address Management

FR-1: The system shall allow authorized users to create and manage administrative addresses.

FR-2: Each address shall reference a parent administrative unit to establish hierarchical relationships.

FR-3: The system shall prevent the creation of circular hierarchy relationships.

FR-4: Users shall access only data within their assigned administrative scope and subordinate units.

3.2 Organization and Store Management

FR-5: The system shall allow authorized users to create and manage organizations within administrative addresses.

FR-6: Each organization shall have at least one store associated with it.

FR-7: Stock shall be managed independently per store with separate inventory ledgers.

3.3 Asset Classification

FR-8: The system shall support Asset Groups identified by 4-digit codes.

FR-9: The system shall support Asset Subgroups identified by 5-digit codes.

FR-10: The system shall support Asset definitions under subgroups with detailed specifications.

FR-11: Asset codes shall be unique across the entire system.

FR-12: Assets shall be categorized as Fixed Asset, Long-term Utility, or Short-term Utility.

3.4 Stock Management

FR-13: The system shall record stock receipt transactions with complete audit trail.

FR-14: The system shall record stock issuance transactions linked to approved requests.

FR-15: The system shall record stock transfer transactions between stores within the same or different organizations.

FR-16: The system shall record stock disposal transactions with appropriate authorization.

FR-17: The system shall prevent stock issuance when available quantity is insufficient.

FR-18: All stock transactions shall be audited with timestamp, user, and transaction details.

3.5 Fixed Asset Management

FR-19: The system shall register fixed assets with a unique Fixed Asset Registration (FAR) number.

FR-20: Fixed assets shall track complete assignment history including user, date, and location.

FR-21: The system shall calculate depreciation of fixed assets based on configurable depreciation methods.

FR-22: Disposal of fixed assets shall require appropriate authorization based on asset value and organizational policy.

3.6 Barcode or QR Asset Tagging

FR-23: The system shall generate unique barcode or QR codes for all fixed assets.

FR-24: Each fixed asset shall be linked to a scannable barcode or QR code containing asset identification information.

FR-25: The system shall support asset lookup and retrieval via barcode/QR scanning.

FR-26: Barcode/QR labels shall be printable in standard formats compatible with common label printers.

FR-27: Scanning a barcode/QR code shall retrieve complete asset profile and current status information.

3.7 Request Workflow

FR-28: Employees shall be able to submit asset requests through the system.

FR-29: Directors and authorized approvers shall be able to approve or reject requests.

FR-30: Only approved requests shall allow stock issuance by storekeepers.

FR-31: Request status shall include the following states: *Waiting*, *Approved*, *Rejected*, *Issued*, *Cancelled*.

3.8 Reporting

FR-32: The system shall generate reports filtered by administrative level.

FR-33: Reports shall support hierarchical aggregation from lower to higher administrative levels.

FR-34: Reports shall allow filtering by date range, organization, asset type, and other relevant criteria.

FR-35: Reports shall be exportable in standard formats including PDF, Excel, and CSV.

4. SYSTEM ACTORS AND ROLE DEFINITIONS

This section provides detailed definitions of all system actors, their categories, scope of authority, and key responsibilities.

4.1 Actor Classification

System actors are categorized into the following types:

- **Operational Actors:** Perform day-to-day transactions and operations
- **Approval Authorities:** Review and approve requests and transactions
- **Administrative Actors:** Configure system settings and manage users
- **Technical Support Actors:** Provide technical assistance and user support

4.2 Actor Definitions

4.2.1 Property Administration Officer (PAO)

Category: Approval Authority
Scope: Regional-level oversight and approval

Key Responsibilities:

- View all assets and detailed asset information across the region
- Generate and print Barcode/QR codes for individual or bulk assets
- Manage stock movement operations (Receive, Issue, Transfer, Dispose)
- View complete bin card history for all stores
- Approve requests requiring PAO-level authorization
- View all transaction records and audit trails
- Execute batch asset operations for efficiency
- Register and manage fixed assets
- Authorize asset disposals

4.2.2 Procurement Directorate Officer (PPO)

Category: Approval Authority
Scope: Procurement-related approvals

Key Responsibilities:

- View all assets within scope of authority
- Generate and print barcodes for assets
- Manage stock flow related to procurement activities
- Approve procurement-related requests
- Modify or remove user requests when necessary

- Send or cancel requests based on procurement decisions
- Forward approved requests to store for fulfillment
- Coordinate with suppliers and vendors

4.2.3 Store Keeper

Category: Operational Actor
Scope: Store-level inventory management

Key Responsibilities:

- Maintain accurate store inventory records
- Receive stock from suppliers or other stores
- Issue stock based on approved requests
- Transfer stock between stores as authorized
- Maintain bin card records for all transactions
- Process approved requests in a timely manner
- Execute batch operations for efficiency
- Conduct periodic physical inventory counts
- Report discrepancies and stock issues

4.2.4 Stock Clerk

Category: Operational Actor
Scope: Assist storekeeper with inventory operations

Key Responsibilities:

- Assist Storekeeper with daily operations
- Manage stock transactions under supervision
- Maintain bin card records
- View transaction records and history
- Prepare stock reports
- Assist with physical inventory counts

Limitations:

- No approval authority
- Cannot authorize transactions independently

4.2.5 Public Body Staff (Employee)

Category: Operational Actor
Scope: Request submission and tracking

Key Responsibilities:

- Create asset requests for required items
- Modify requests before approval
- Cancel requests when no longer needed
- Track status of submitted requests
- View personal request history

Limitations:

- Cannot manage stock directly
- Cannot approve requests
- Cannot access other users' requests

4.2.6 Work Unit Head

Category: Approval Authority
Scope: Work unit level approvals

Key Responsibilities:

- Approve or reject asset requests from subordinate staff
- Approve transfers of assets between staff members
- Approve issue and purchase requests within authority limits
- Monitor asset utilization within work unit
- Review work unit asset reports

Limitations:

- Cannot manage stock directly
- Cannot perform store operations

4.2.7 System Administrator

Category: Administrative Actor
Scope: System-wide configuration and user management

Key Responsibilities:

- Manage user accounts (create, modify, deactivate)
- Manage groups and roles
- Manage permissions and access control
- Configure stores and organizational units
- Configure master data (asset groups, manufacturers, etc.)
- Monitor system performance and usage
- Perform system backups and maintenance

Limitations:

- No operational stock authority
- Cannot approve requests or perform transactions

4.2.8 ICT Support

Category: Technical Support Actor
Scope: Technical assistance and user support

Key Responsibilities:

- Create and modify user accounts as directed
- View groups and permissions for troubleshooting
- Assist users with access issues
- Provide technical support for system usage
- Report system issues to administrators

Limitations:

- Cannot delete permissions
- Cannot perform stock operations
- Cannot modify system configuration

5. USE CASES

This section describes the primary use cases for the Procurement and Property Management System. Each use case follows a standard format including actors, preconditions, main flow, and postconditions.

UC-01: Submit Asset Request

Actor: Employee

Precondition:

- User is authenticated and authorized
- User has access to request submission functionality

Main Flow:

1. Employee navigates to asset request form
2. Employee selects required asset from available catalog
3. Employee enters required quantity
4. Employee provides justification for request
5. Employee submits request
6. System validates request data
7. System records request with status = *Waiting*
8. System sends notification to approver

Postcondition:

- Request is created and awaiting approval
- Employee can track request status

Alternative Flow:

- If validation fails, system displays error message and allows correction

---

UC-02: Modify Request

Actor: Employee

Precondition:

- User is authenticated
- Request exists with status = *Waiting*
- User is the request originator

Main Flow:

1. Employee navigates to pending requests
2. Employee selects request to modify
3. System displays request details
4. Employee modifies asset selection or quantity
5. Employee saves changes
6. System validates modified data
7. System updates request record
8. System maintains audit trail of modifications

Postcondition:

- Request is updated with new information
- Request remains in *Waiting* status

Alternative Flow:

- If request has been approved, system prevents modification and displays message

---

UC-03: Cancel Request

Actor: Employee

Precondition:

- User is authenticated
- Request exists and is not fully approved or issued
- User is the request originator

Main Flow:

1. Employee navigates to active requests
2. Employee selects request to cancel
3. Employee clicks Cancel button
4. System prompts for cancellation confirmation
5. Employee confirms cancellation
6. System updates request status = *Cancelled*
7. System sends notification to relevant parties

Postcondition:

- Request is cancelled and cannot be processed further

UC-04: Work Unit ApprovalUC-04: Work Unit Approval

Actor: Work Unit Head

Precondition:

- User is authenticated with Work Unit Head role
- Request exists with status = *Waiting*
- Request is within user's approval scope

Main Flow:

1. Work Unit Head navigates to pending approvals
2. System displays requests awaiting approval
3. Work Unit Head reviews request details
4. Work Unit Head selects Approve or Reject action
5. Work Unit Head provides comments (optional)
6. Work Unit Head confirms decision
7. System updates request status accordingly
8. System sends notification to requester and next approver

Postcondition:

- Request status is updated to *Work Unit Approved* or *Rejected*
- If approved, request moves to next approval level

Alternative Flow:

- If rejected, request status = *Rejected* and workflow terminates

---

UC-05: PPO Approval

Actor: Procurement Directorate Officer (PPO)

Precondition:

- User is authenticated with PPO role
- Request has Work Unit approval
- Request requires PPO-level approval

Main Flow:

1. PPO navigates to pending approvals
2. System displays requests awaiting PPO approval
3. PPO reviews request details and justification

4. PPO verifies budget availability (if applicable)
5. PPO selects Approve or Reject action
6. PPO provides comments
7. PPO confirms decision
8. System updates request status
9. System sends notification to relevant parties

Postcondition:

- Request status is updated to *PPO Approved* or *Rejected*
- If approved, request moves to PAO approval (if required) or to store for fulfillment

---

UC-06: PAO Approval

Actor: Property Administration Officer (PAO)

Precondition:

- User is authenticated with PAO role
- Request has PPO approval
- Request requires PAO-level approval

Main Flow:

1. PAO navigates to pending approvals
2. System displays requests awaiting PAO approval
3. PAO reviews complete request history and approvals
4. PAO verifies compliance with policies
5. PAO selects Approve or Reject action
6. PAO provides final comments
7. PAO confirms decision
8. System updates request status to final approval state
9. System notifies storekeeper to fulfill request

Postcondition:

- Request status is updated to *Fully Approved* or *Rejected*
- If approved, request is ready for stock issuance

UC-07: Receive Stock

Actor: Store Keeper

Precondition:

- User is authenticated with Store Keeper role
- Valid procurement order or approved receipt exists
- Supplier delivery has been verified

Main Flow:

1. Store Keeper navigates to stock receipt function
2. Store Keeper enters or selects procurement order reference
3. Store Keeper enters asset details (type, quantity, unit price)
4. Store Keeper enters supplier information
5. Store Keeper enters receipt date and reference number
6. Store Keeper confirms receipt
7. System validates data
8. System updates stock ledger with receipt transaction
9. System increases running balance
10. System generates receipt document (Model 19)
11. System updates bin card

Postcondition:

- Stock quantity is increased
- Receipt transaction is recorded
- Bin card is updated

UC-08: Issue Stock

Actor: Store Keeper

Precondition:

- User is authenticated with Store Keeper role
- Request is fully approved
- Sufficient stock is available

Main Flow:

1. Store Keeper navigates to approved requests
2. System displays requests ready for fulfillment
3. Store Keeper selects request to fulfill
4. System displays request details and required items
5. System validates stock availability
6. Store Keeper confirms issuance

7. System deducts stock quantity
8. System updates stock ledger with issue transaction
9. System decreases running balance
10. System generates issue document (Model 22)
11. System updates bin card
12. System updates request status = *Issued*
13. System sends notification to requester

Postcondition:

- Stock quantity is decreased
- Issue transaction is recorded
- Request is fulfilled
- Requester is notified

Alternative Flow:

- If insufficient stock, system displays error and prevents issuance

---

UC-09: Transfer Stock

Actor: Store Keeper

Precondition:

- User is authenticated with Store Keeper role
- Sufficient stock is available in source store
- Transfer is authorized

Main Flow:

1. Store Keeper navigates to stock transfer function
2. Store Keeper selects source store
3. Store Keeper selects destination store
4. Store Keeper selects asset to transfer
5. Store Keeper enters quantity to transfer
6. Store Keeper provides transfer reason
7. System validates stock availability
8. Store Keeper confirms transfer
9. System deducts stock from source store atomically
10. System adds stock to destination store atomically
11. System records transfer transaction in both stores
12. System updates bin cards for both stores

13. System generates transfer document

Postcondition:

- Stock is transferred between stores
- Both store ledgers are updated
- Transfer is documented

Alternative Flow:

- If insufficient stock, system displays error and prevents transfer
- If transaction fails, system rolls back all changes

UC-10: Dispose Asset

Actor: Property Administration Officer (PAO) / Store Keeper

Precondition:

- User is authenticated with appropriate role
- Asset exists and is marked eligible for disposal
- Disposal is authorized per policy

Main Flow:

1. User navigates to asset disposal function
2. User searches for and selects asset to dispose
3. System displays asset details and history
4. User enters disposal reason and method
5. User enters disposal date and reference
6. User uploads supporting documentation (if required)
7. User confirms disposal
8. System validates authorization
9. System updates asset status = *Disposed*
10. System updates stock ledger
11. System generates disposal document
12. System updates bin card

Postcondition:

- Asset is marked as disposed
- Stock quantity is adjusted
- Disposal is documented

UC-11: Maintain Bin Card

Actor: Store Keeper / Stock Clerk

Precondition:

- Stock transaction occurs (Receive, Issue, Transfer, or Dispose)

Main Flow:

1. User performs stock transaction
2. System automatically records transaction in bin card
3. System updates running balance
4. System timestamps transaction
5. System records user who performed transaction
6. System displays updated bin card
7. User can view complete transaction history

Postcondition:

- Bin card is updated with transaction
- Running balance reflects current stock level
- Transaction history is maintained

UC-12: Generate Barcode / QR Code

Actor: Property Administration Officer (PAO) / Procurement Directorate Officer (PPO) / Store Keeper

Precondition:

- User is authenticated with appropriate role
- Asset is registered in system

Main Flow:

1. User navigates to barcode/QR generation function
2. User selects single asset or bulk generation option
3. If single: User searches for and selects asset
4. If bulk: User selects multiple assets or asset group
5. User selects code format (Barcode or QR Code)
6. User clicks Generate button
7. System generates unique code(s) for selected asset(s)
8. System associates code(s) with asset record(s)
9. System displays preview of generated code(s)

10. User saves printable file (PDF format)
11. User prints labels using standard label printer

Postcondition:

- Barcode/QR code is generated and associated with asset
- Printable label file is available
- Asset can be scanned for identification

UC-13: Register Fixed Asset (FAR Registration)

Actor: Property Administration Officer (PAO)

Precondition:

- User is authenticated with PAO role
- Asset has been received into inventory
- Asset qualifies as fixed asset per policy

Main Flow:

1. PAO navigates to fixed asset registration function
2. PAO enters asset details (description, specifications, cost)
3. PAO enters acquisition date and supplier information
4. PAO selects asset category and subcategory
5. PAO enters depreciation method and useful life
6. System generates unique FAR number
7. System generates Barcode/QR code automatically
8. PAO assigns asset to location or user (optional)
9. PAO confirms registration
10. System saves fixed asset record
11. System updates asset status = *Active*
12. System generates FAR certificate

Postcondition:

- Fixed asset is registered with unique FAR number
- Asset has associated Barcode/QR code
- Asset is ready for assignment and tracking

UC-14: Scan Asset Barcode / QR

Actor: Store Keeper / Auditor / Property Officer

Precondition:

- User is authenticated
- Asset has assigned Barcode/QR code
- Scanning device is available and configured

Main Flow:

1. User navigates to asset scan function
2. User activates scanner (camera or barcode reader)
3. User scans asset Barcode/QR code
4. System decodes scanned information
5. System retrieves asset record from database
6. System displays complete asset profile including:
    - Asset identification and description
    - Current status and location
    - Assignment history
    - Transaction history
    - Maintenance records (if applicable)
7. User can perform additional actions (update, transfer, etc.)

Postcondition:

- Asset information is retrieved and displayed
- User can verify asset details

Alternative Flow:

- If code is invalid or not found, system displays error message

---

UC-15: Manage Users

Actor: System Administrator

Precondition:

- User is authenticated with System Administrator role

Main Flow:

1. Administrator navigates to user management function
2. Administrator selects action: Create, Update, or Delete user
3. **If Create:**
    - Administrator enters user details (name, email, username)
    - Administrator assigns role/group

    - Administrator assigns administrative scope
    - System generates temporary password
    - System creates user account
4. **If Update:**
    - Administrator searches for and selects user
    - Administrator modifies user details or role
    - System validates changes
    - System updates user record
5. **If Delete:**
    - Administrator searches for and selects user
    - System prompts for confirmation
    - Administrator confirms deletion
    - System deactivates user account
6. System updates access permissions based on role
7. System logs administrative action

Postcondition:

- User account is created, updated, or deactivated
- Access permissions are updated accordingly
- Action is logged for audit

UC-16: Configure Master Data

Actor: System Administrator

Precondition:

- User is authenticated with System Administrator role

Main Flow:

1. Administrator navigates to master data configuration
2. Administrator selects master data type:
    - Asset Group
    - Asset Subgroup
    - Store
    - Organization
    - Manufacturer
    - Supplier
    - Unit of Measure

    - Other configuration data
3. Administrator selects action: Add, Update, or Delete record
4. **If Add:**
    - Administrator enters required information
    - System validates data
    - System saves new record
5. **If Update:**
    - Administrator selects existing record
    - Administrator modifies information
    - System validates changes
    - System updates record
6. **If Delete:**
    - Administrator selects record to delete
    - System checks for dependencies
    - If no dependencies, system deletes record
    - If dependencies exist, system prevents deletion and displays message
7. System logs configuration change

Postcondition:

- Master data is created, updated, or deleted
- Configuration change is logged
- System reflects updated configuration

---

6. NON-FUNCTIONAL REQUIREMENTS

This section specifies the non-functional requirements that define system quality attributes and constraints.

6.1 Performance

NFR-1: The system shall support at least 500 concurrent users without performance degradation.

NFR-2: Page load time shall not exceed 3 seconds under normal network conditions.

NFR-3: Database queries shall return results within 2 seconds for 95% of requests.

NFR-4: Report generation shall complete within 30 seconds for standard reports.

NFR-5: Barcode/QR scanning shall provide response within 1 second.

6.2 Security

NFR-6: The system shall implement role-based access control (RBAC) for all functions.

NFR-7: All user passwords shall be encrypted using industry-standard hashing algorithms.

NFR-8: The system shall enforce password complexity requirements.

NFR-9: User sessions shall be timed out after 30 minutes of inactivity.

NFR-10: All transactions shall be logged with user identification and timestamp.

NFR-11: The system shall protect against SQL injection and cross-site scripting (XSS) attacks.

NFR-12: Data transmission shall be encrypted using HTTPS/TLS protocols.

6.3 Availability

NFR-13: The system shall maintain 99% uptime during business hours (8:00 AM - 5:00 PM).

NFR-14: Planned maintenance shall be scheduled during off-peak hours with advance notice.

NFR-15: The system shall implement automated backup procedures with daily incremental and weekly full backups.

NFR-16: System recovery time objective (RTO) shall not exceed 4 hours.

NFR-17: Data recovery point objective (RPO) shall not exceed 24 hours.

6.4 Scalability

NFR-18: The system architecture shall support horizontal scaling to accommodate growth.

NFR-19: The database shall support partitioning for large transaction tables.

NFR-20: The system shall handle up to 10,000 asset records per organization without performance impact.

NFR-21: The system shall support the addition of new administrative units without system redesign.

7. SYSTEM ARCHITECTURE OVERVIEW

This section provides a high-level overview of the system architecture and technical design approach.

7.1 Architectural Style

The Property Management System employs a **Microservice Architecture**. This approach provides:

- Clear separation of concerns
- Maintainability and testability
- Flexibility for future enhancements
- Simplified deployment and operations

7.2 Technology Stack

Database: PostgreSQL relational database management system

Key Database Features:

- Partitioned transaction tables for performance optimization
- Running balance table for efficient stock queries
- Materialized views for complex reporting requirements
- Referential integrity enforcement
- Transaction support for data consistency

7.3 Data Management Strategy

Transaction Management:

- All stock transactions are recorded in partitioned tables
- Running balance is maintained in a separate optimized table
- Historical data is preserved for audit and compliance

Reporting:

- Materialized views provide pre-aggregated data for common reports
- Views are refreshed on a scheduled basis
- Ad-hoc queries are supported for custom reporting needs

Data Isolation:

- Logical data isolation based on administrative hierarchy
- Row-level security enforces access boundaries
- Users can only access data within their scope of authority

7.4 Integration Considerations

The system architecture allows for future integration with:

- Financial management systems
- Procurement systems
- Human resource management systems
- External reporting platforms

Integration points shall be designed using standard APIs and data exchange formats.